

Variational Perturbation Theory for Strongly Interacting Quantum Systems

Benchmarked with the Anharmonic Oscillator

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Introduction

In many-body physics, propagators are written as a perturbative series of Feynman diagrams. These are often evaluated either by stopping at a finite order or by partial resummation techniques in order to avoid divergences in the strong coupling limit.

Variational perturbation theory (VPT) supplies a perturbative series capable of highly accurate non-perturbative strong coupling calculations by introducing a variational parameter to the usual method.

The observable x^4 alludes to particle interactions if x is treated as a creation/annihilation operator. A useful benchmark then is to consider a VPT formulation on the 1D anharmonic oscillator like so:

$$\hat{H} = \underbrace{\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2}_{\hat{H}_0} + \underbrace{\frac{1}{2}m\Omega^2\hat{x}^2 - \frac{1}{2}m\Omega^2\hat{x}^2 + g\hat{x}^4}_{\hat{H}_1} \quad (1)$$

To find Ω , we use the principle of minimal sensitivity (PMS) on the expansion, which is the statement that the perturbative expansion of some observable A depends as little as possible on Ω , and in the ideal case:

$$\left. \frac{dA^{(n)}}{d\Omega} \right|_{\Omega^*} = 0 \quad (2)$$

Once a value for Ω^* is found, simply plug it back into the perturbative expansion to find the prediction for that observable. This procedure is simplest for energies but can be done for other observables as we will see for x^2 .

Results

The (numerically) exact energies can be found by solving the Hamiltonian's eigenvalue problem numerically in the oscillator basis. The difference between various approximation schemes relative to the exact energies is shown in Figure 1.

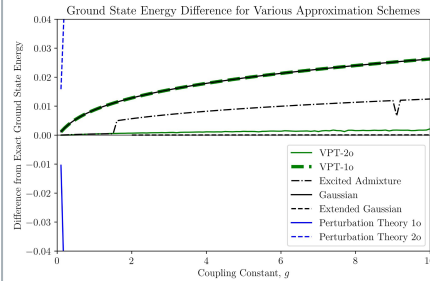


Figure 1. Ground State Energy Approximations (Constants set to 1)

The black lines in Figure 1 represent a variational method. Based on previous discussion, Figure 1 suggests that VPT is a viable technique for strongly interacting many-body systems.

Another observable of interest is $\langle x^2 \rangle$, which alludes to particle density under $x \rightarrow \psi$. A comparison between the various VPT procedures using (2) is supplied in Figures 2, 3.

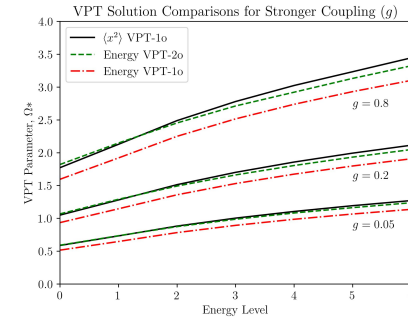


Figure 2. VPT Parameter Solutions at Strong Coupling

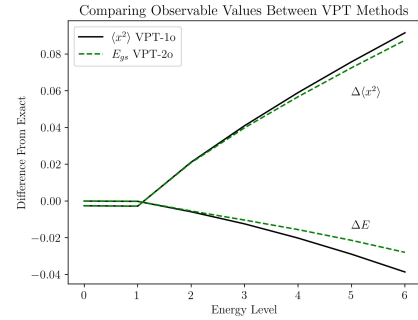


Figure 3. VPT Observable Predictions

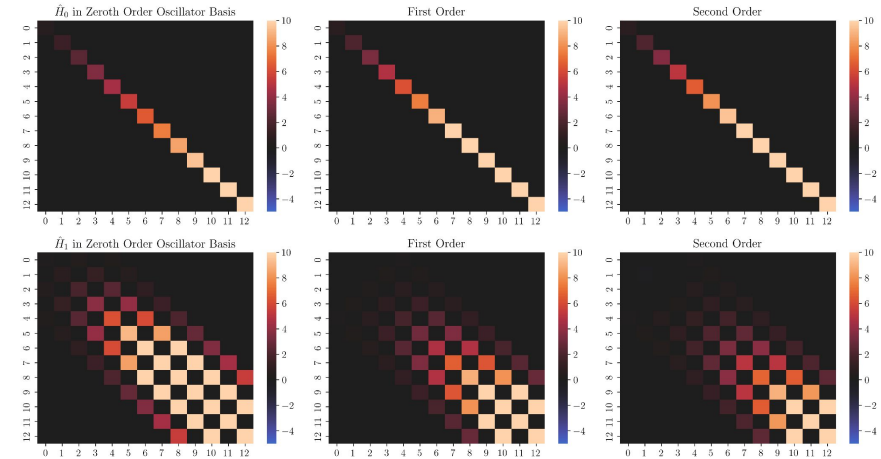


Figure 4. Hamiltonian Heatmaps in Oscillator Basis

Why does it work?

Eq. (1) suggests that VPT “absorbs” some of the perturbative effects into the unperturbed Hamiltonian so that the perturbation is weaker in comparison. In introducing the VPT parameter we are considering a basis where the perturbation has weaker expression, as seen in Figure 4 for the ground state energy.

Future Work

Future work for VPT lies in making the jump to a true many body system, expressing it in a quantum field theoretical framework to see if the analogies hold.

Acknowledgements

I am very grateful to the Department of Physics at The Ohio State University and the Bunny C. Clark Student Support Fund for this opportunity.